

EXPERIMENT NO: 23

ACCESS AN INTERNAL ADC AND DISPLAY THE BINARY OUTPUT IN LEDs IN LPC2148 KIT

AIM

To write and execute C program for accessing an internal ADC and display the binary output in LEDs in LPC2148 kit.

OBJECTIVES

- Initialize and access the internal ADC channel AD0.1 of the LPC2148.
- Convert the analog input into a digital ADC value and reduce it to an 8-bit LED value.
- Display the converted binary value on LEDs P0.8 to P0.15.

APPARATUS / SOFTWARE REQUIRED

- Keil uVision5 Software
- Philips Flash Programmer
- LPC 2148 kit

PROGRAM

```
#include <LPC214X.H>

#define LEDS 0xFF<<8          //LED => P0.8 to P0.15

////////////////////////////////////
/*--- ADC Signal Declaration */
////////////////////////////////////
#define AD0_1 1<<24
#define CLK_DIV 1<<8
#define PDN 1<<21
#define SOC 1<<24
#define BURST 1<<16
#define DONE 1<<31

/*-----*/
//Delay Program
//Input - delay value in milli seconds
void delay(unsigned int k)
{
    unsigned int i,j;
    for (j=0; j<k; j++)
        for(i = 0; i<=800; i++);
}

/*-----*/
void adc_init()
{
    unsigned long int ADC_CH;
    ADC_CH = 0 | 1 << 1;          //Channel AD0.1
    AD0CR = SOC | PDN | CLK_DIV | ADC_CH | BURST;
}

/*-----*/
unsigned int adc_read(unsigned char channel)
{
    unsigned int aval;
    unsigned long int val;

    if (channel == 1) val = AD0DR1;
    else if (channel == 2) val = AD0DR2;
    else if (channel == 3) val = AD0DR3;

    val = val >> 6;
    val = val & 0x3FF;
    aval = val;
    return (aval);
}
```

```

/*-----*/
/*----Main Program-----*/
int main(void)
{
    unsigned int tp1;

    IODIR0 = LEDS;           //Configure Port0 as output Port
    PINSEL0 = 0;             //Configure Port0 as General Purpose IO
    PINSEL1 = 0 | AD0_1;     //Enable AD0.1
    adc_init();              //Initialise on-chip ADC

    do
    {
        tp1 = adc_read(1);   //Channel AD0.1
        tp1 = tp1 >> 2;      //ADC 10 bit but LED 8 bit, truncate LSB 2 bits
        IOSET0 = LEDS;       //Switch OFF all LEDs
        IOCLR0 = tp1 << 8;    //Set value
        delay(1000);
    }while(1);
}

```

OUTPUT

The potentiometer knob was adjusted to generate Analog input and Digital display is observed.

RESULT

Thus C program was written and executed for accessing an internal ADC and display the binary output in LEDs in LPC2148 kit.